

Nafiul Islam Naim

B.Sc. (Honours), 4th-Year Student
Department of Applied Mathematics
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🎓 Education

Bachelor of Science Oct 2022 – Expected Dec 2026

Department of Applied Mathematics, University of Dhaka, Dhaka, Bangladesh

CGPA: 3.00/4.00 (through 3rd year)

Undergraduate Project : “ Principal Component Analysis and Its Applications ” — Developed a PCA-Based machine learning framework for predicting annual Heat Wave Days (HWDs) across Bangladesh using ERA5 reanalysis climate data.

Supervisor: [Shuchi Chaki](#)

🔬 Research Interests

My research interests focus on **Learning and Foundation Models**, **Computer Vision and Multi-modal AI**, **AI Agents and Autonomous Systems**, **Robotics and Physical AI**, **AI for Climate and Earth System Science**, **Data Science and Statistical Learning**, and **Optimization and Scientific Computing**. I am looking to work with complex data and applying computational and AI techniques to find solutions to real-world problems, specifically finding efficient, resource-limited, and interpretable solutions.

📖 Research Project (Ongoing)

BD-HeatPred: A Multi-Source Daily Predictor Dataset for Heatwave Research in Bangladesh, February–June, 1991–2025

🔗 [project page](#)

- Developing a region-specific dataset for Bangladesh by integrating 35 atmospheric, land-surface, and global teleconnection predictors across five regions: Dhaka, Khulna, Rajshahi, Rangpur, and the Rangamati–Bandarban–Khagrachari (RBK).

BD-HeatPred-Based Heatwave Onset Prediction and Regional Driver Analysis in Bangladesh

- Heatwave onset prediction in Bangladesh using the BD-HeatPred dataset, with regional driver analysis to identify the most influential predictors across five regions :Dhaka, Khulna, Rajshahi, Rangpur, and the Rangamati–Bandarban–Khagrachari (RBK).

</> Projects

Principal Component Analysis and Its Applications

🔗 [GitHub](#)

- Developed an interpretable PCA-based machine learning framework to predict annual Heat Wave Days (HWDs) across Bangladesh's eight administrative divisions using ERA5 atmospheric data .
- Evaluated the effectiveness of PCA for dimensionality reduction and prediction performance, and applied SHAP-based explainable AI to identify dominant atmospheric drivers associated with heat-wave occurrence across Bangladesh.

Tools: Python, Pandas, NumPy, Scikit-learn, PCA, mRMR, SHAP, ERA5

Technical skills

Programming	Python, C, C++, Dart , MATLAB
Data Analysis & Visualization	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
ML & Deep Learning	PyTorch, Keras, Explainable AI (SHAP), PCA, mRMR
Computer Vision	Ultralytics YOLO, OpenCV, Roboflow
Climate & Geospatial Analysis	ERA5/ERA5-Land, NetCDF, Xarray, GeoPandas, GIS, Spatial Analysis, Grid-Based Analysis, Remote Sensing (NDVI, NDWI), Google Earth Engine(GEE), Climate Anomaly & Teleconnection Analysis(SST, ENSO, IOD, MJO, OLR)
Web & Application Development	HTML, CSS, Flutter
Tools	Git, GitHub, VS Code, PyCharm, LaTeX, Google Colab, Jupyter Notebook, Canva

Standardized Tests

I plan to take the **International English Language Testing System (IELTS)** and the **Graduate Record Examination (GRE)** soon.

Referees

- **Shuchi Chaki**

Graduate Teaching Assistant

North Carolina State University

Email: shuchi@du.ac.bd

Lecturer (on study leave)

Department of Applied Mathematics, University of Dhaka, Dhaka, Bangladesh

- **Kawser Ahammed**

PhD Student (fifth year)

Department of Electrical and Computer Engineering

Old Dominion University, Virginia, USA

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[Website](#)

Assistant Professor (on study leave)

Department of Electrical and Electronic Engineering (EEE),

[Jatiya Kabi Kazi Nazrul Islam University, Trishal, Mymensingh, Bangladesh](#)